gases is produced and this exerts a very high pressure force on the top of the piston. Due to this high pressure force, the piston moves downward and rotates the crankshaft and does useful work.

During this stroke the inlet port is covered by the piston and the new charge is compressed in the crankcase due to the downward movement of piston.

Further downward movement of the piston uncovers first the exhaust and the exhaust starts through the exhaust port.

Further downward movement of the piston uncovers port the transfer port and the charge through it is forced into the cylinder.

The charge strikes the deflector on the piston crown, rises to the top of the cylinder and pushes out most of the exhaust gases.

The piston is now at BDC position. The cylinder is completely filled with the fresh charge but it is somewhat diluted with the exhaust gases.

Finally the cycle event is then repeated and the power stroke is obtained for the every single revolution of the crankshaft.

Reason for less power in two stroke engine

- The fresh air fuel charge is mixed with exhaust gas and sent out before combustion
- The charge is diluted by the burnt gases due to incomplete scavenging.
- Combustion is improper as lubrication oil is mixed with fresh charge
- Volumetric efficiency of the engine is low.

Advantages:

- Smoother in operation
- It is simpler in construction and mechanism.

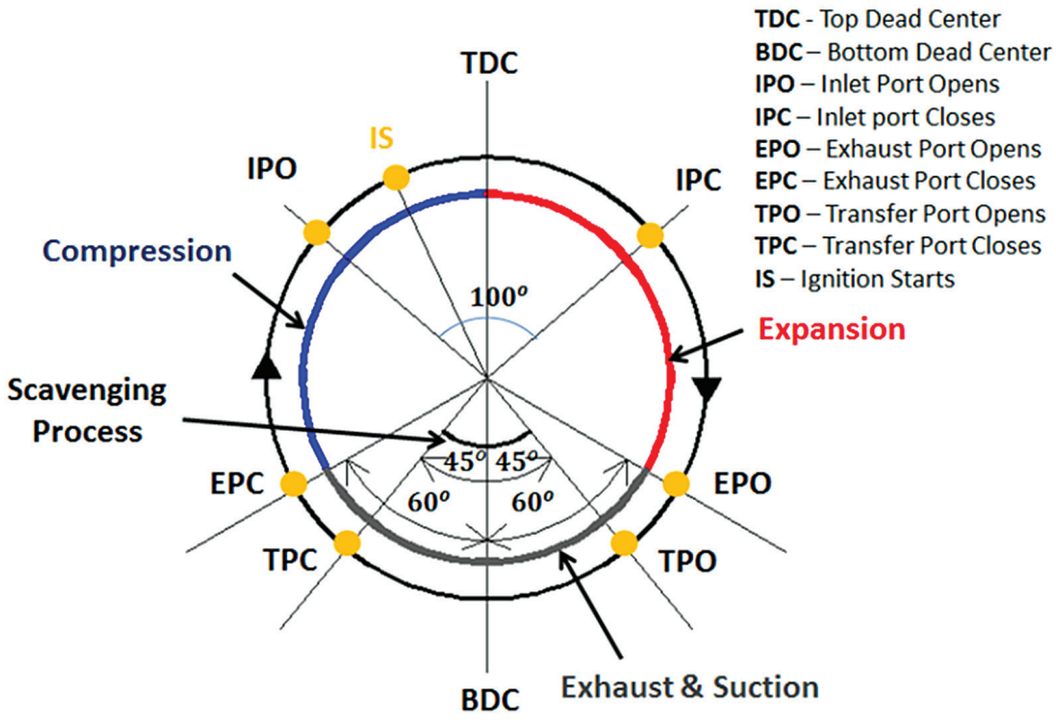


Figure 5.5(c) Port Timing diagram of Two Stroke Petrol Engine

AM_Unit_05.indd 103 12/20/2019 5:26:39 PM